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Agentic AI for Lead Generation

Partner companies for the chair

Tobias Weiß · Guest Lecture · Professur Marketing & Verkaufsmanagement

Thursday, 22 October 2026 · 15:15–16:00
within "Advanced Issues in Marketing and Sales" (Prof. Haas)

45 minutes · BWL Master students · WS 2026/27 edition

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Why this session exists

- The chair partners with companies: funding, data, cases
- Finding + qualifying partners is manual, slow, unscalable
- Agentic AI can research, score and rank prospects
- **Demo output = evidence for "Sales Strategy and Execution" (SS 2027)**

> Nobody wins from another lecture about chatbots.

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What you will be able to do

By the end…	You can…
Understand	Explain what "agentic" means, and why it is not a chatbot
Operate	Read how a harness (pi) gives an AI tools, rules and a task
Assess	Judge an agentic output: checked? citable? repeatable?
Apply	Run the exercise yourself: ingest + rate leads for the chair (repo → rubric → validator)
Context	Say which context an agent can't know — and why that keeps you in charge

> Model = brain. Harness = hands. Workflow = muscles. **Context = memory — the part only you can supply.**

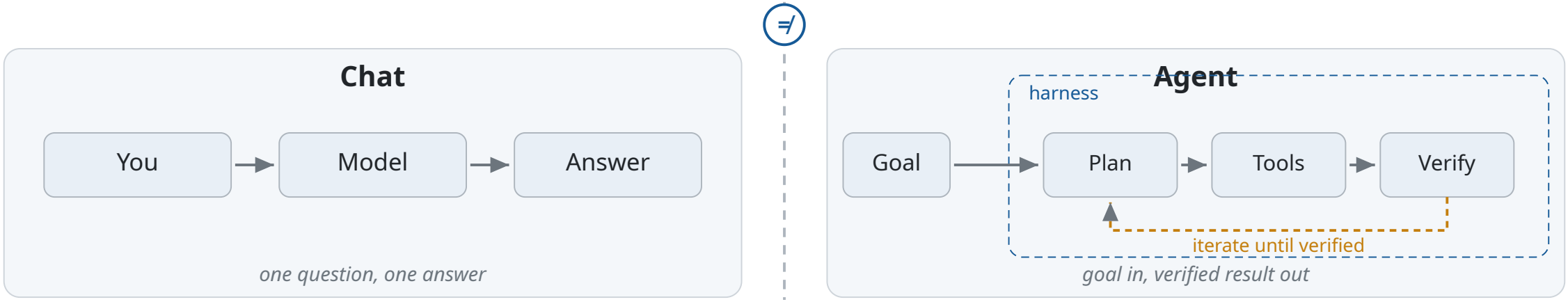
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Roadmap — 35 min + 10 min Q&A

Block	Content	Time
1 · Foundations	agentic ≠ chat · model · harness · loop · providers · git & the repo	~10 min
2 · Method	contracts · the lead-gen loop · rating formulas · evidence	~8 min
3 · Walkthrough	live demo: repo → agent → validator	~13 min
4 · Reflection	take-aways · resources	~4 min
5 · Q&A	your questions · discussion	10 min

> The walkthrough is the heart — theory is fuel for it. **10 minutes are reserved for your questions.**

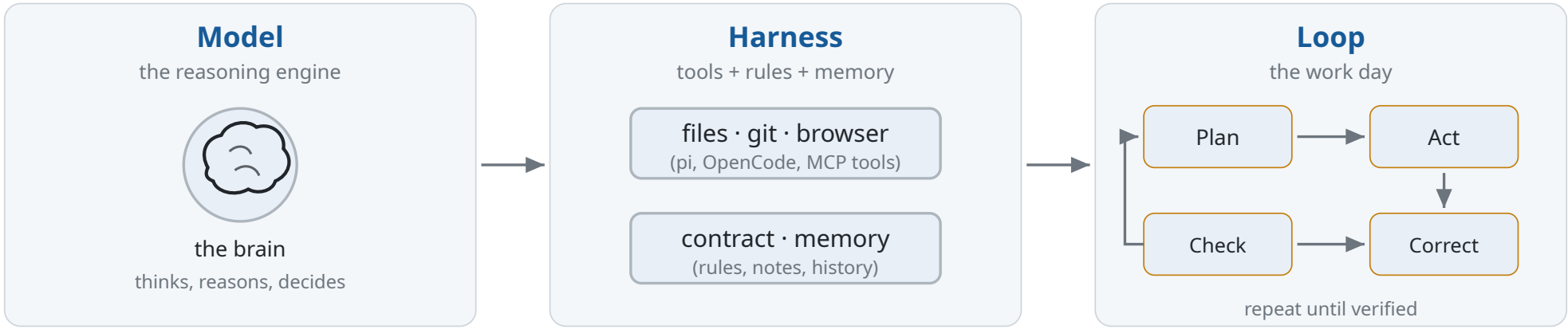
Agentic ≠ chat



> "Agentic" is a working environment — not a better chat.

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The parts: model · harness · loop



Harnesses (2026): pi, OpenCode, Claude Code — same model, different results. The harness decides.

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Key AI providers — what you can use as a student

Provider / offer	What you get	Where
JLU HRZ API-Service ⭐	LLMs for all JLU members: free local Qwen models (256K–1M ctx) + commercial, works with OpenCode / Claude Code	uni-giessen.de → HRZ → KI → API-Service · access via **ki@uni-giessen.de**
Google AI student plan	1 year Google AI **Plus** free · Gemini, higher limits, 400 GB	one.google.com/ai-student — verify via SheerID (student status)
OpenRouter	one API key → many models (OpenAI, Anthropic, Google, open) · free models + pay-as-you-go	openrouter.ai
Harnesses	the tools from today's demo — **OpenCode**, pi — work with any key above	opencode.ai · pi.dev
Local models (Ollama)	free, run on your own laptop, no account, private	ollama.com

> Start with **HRZ API** (free for JLU) · Google deal expires **Dec 31, 2026** · campus DNS can't resolve `opencode.ai` ?
> → resolver `1.1.1.1` / `8.8.8.8` or hosts entry `172.65.90.20 opencode.ai` — how-to: one-pager

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The workspace: git and the repo itself

git = a lab notebook that never forgets

This repo — `marketing-research`, public	The numbers
The **corpus**: >14,000 papers in one `papers.yaml` — title · authors · abstract · venue · URL	**>150 commits** since Aug 2026 · 1 researcher + AI agents · >90 last month
The **machinery**: taxonomy · discovery scripts · **executable validators**	generated files are **never hand-edited**
Your exercise: `leads/` — same rules, same history	every lead a commit · every claim a URL

`github.com/tobias-weiss-ai-xr/marketing-research`

> If it isn't in the repo, it didn't happen. If it is, you can check it — line by line, back to day one.

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Why now — the evidence inside the research corpus

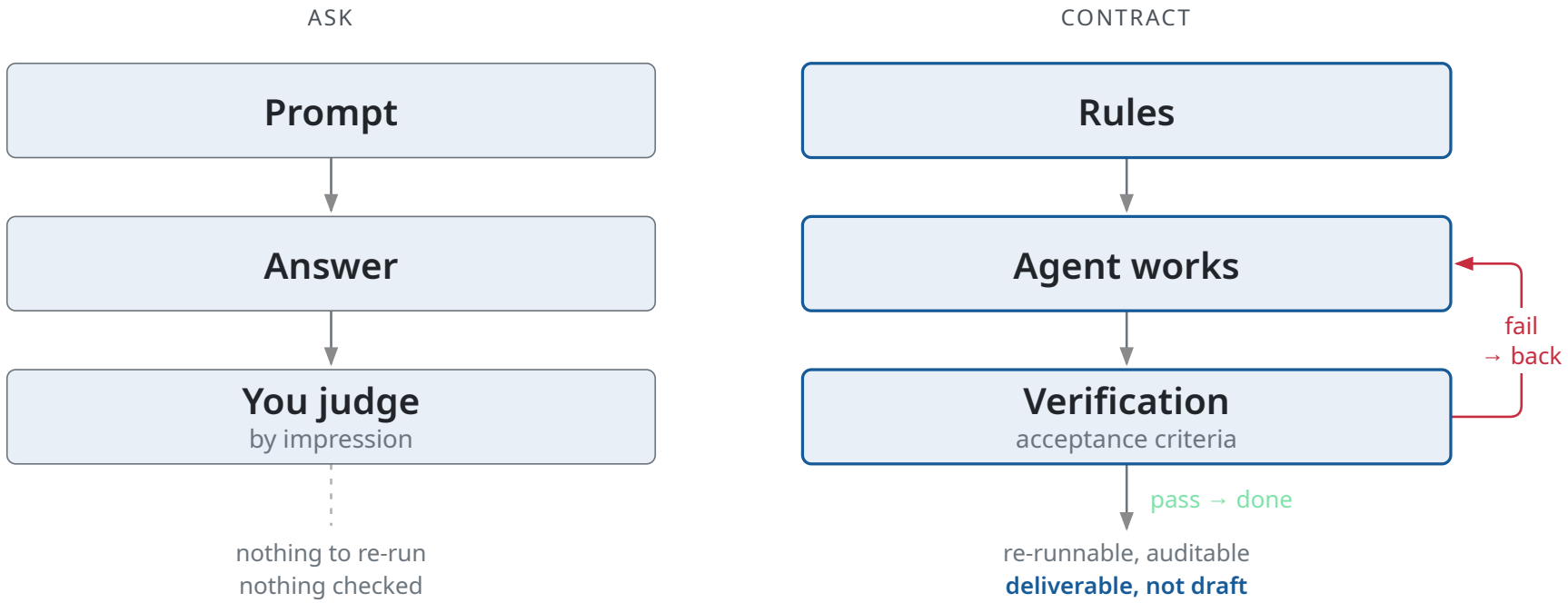
The chair's open corpus, public on GitHub: **>14,000 marketing papers — >8,700 published in 2026 alone**

Signal	Count
Papers mentioning **agentic AI**	>100 — >90 of them from 2026
Papers on **lead generation**	>30
Both — strict (agentic × lead-gen)	**≥ 2**
Both — broader (agent-aware × lead-ops)	**≥ 5**
Papers on **context-aware** marketing	>20

> Two rounds of OpenAlex ingestion took the strict intersection from **1** to **≥ 2**, the broader set from **3** to **≥ 5**. The field just opened — claim the gap while it's a gap.

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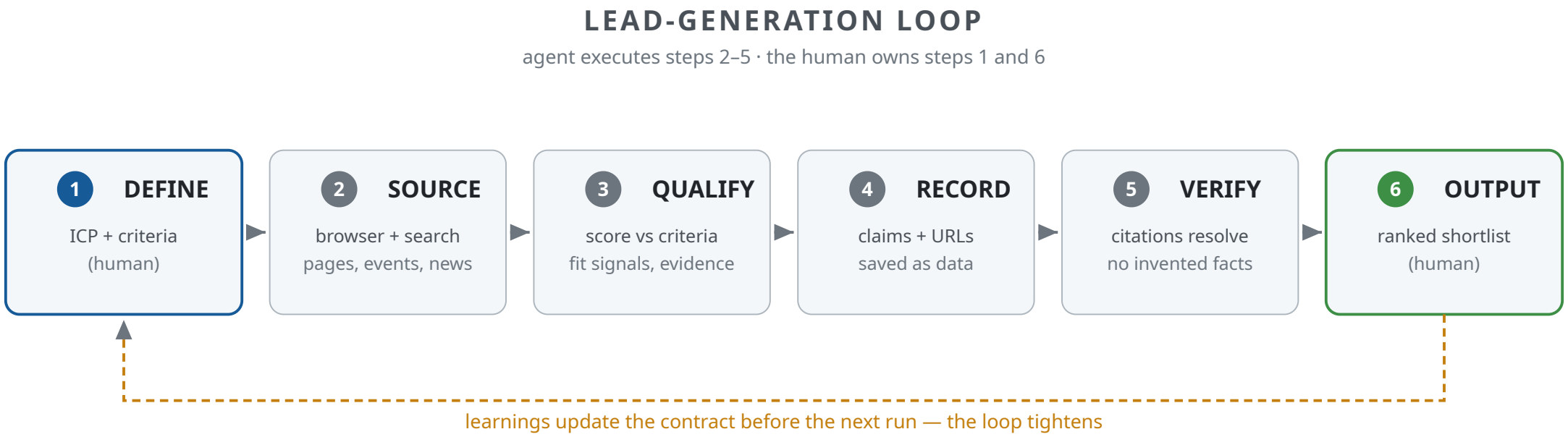
How to talk to the machine: prompts vs contracts



Prompt = asks a chat · Contract = runs an agent · **Context = what the machine can't know**

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A lead-generation loop as an agentic workflow



Six steps — the agent executes 2–5, you own definition and evaluation. **Those two boxes are where context lives.**

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Define the hunt: scoring the sector

Lead generation starts with a **human** decision: *which sector do we hunt?*

Criterion (1–5)	Mech. & plant engineering	Optics / photonics	Grocery retail
Chair research fit — sales management	5	5	2
Course fit — "Sales Strategy and Execution"	5	5	2
Regional employers (Mittelhessen)	5	5	3
Lead-gen relevance — B2B, long cycles	5	4	2
Demo data availability	4	4	5
Total	**24**	**23**	**14**

Hard guard: sector is out if chair fit < 4 **or** lead-gen relevance < 4 — grocery retail (our first default) fails.

Decision: German mechanical & plant engineering.

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Rate a lead: pick a formula — before the agent runs

Formula	Rule	Trade-off
Weighted sum	`L = Σ wᵢ·sᵢ` · w: fit .35 · region .25 · size .15 · evidence .15 · timing .10	transparent · **compensatory**: a 5 on fit can hide a 0 on evidence
Geometric mean	`L = Π sᵢ^wᵢ`	**non-compensatory**: one zero sinks the lead · harsh to defend
Gated weighted sum ✓	gates first — B2B · region reachable · ≥ 1 verified source — then weighted sum	matches the chair's decision · thresholds = judgement calls

Example (each criterion 0–5): fit 5 · region 3 · size 4 · evidence 4 · timing 2 → `L = 3.90`

> Weights live in the repo (`leads/leads.yaml` , `rubric:`) — **published context, not vibes**. Change them — in a PR, with reasons.

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Now: a live walkthrough

Task: ingest **2 leads** into the chair's repo — `leads/leads.yaml` — with:

- scores per criterion + the weighted sum, gates applied, ranked
- named source URL per claim (citable, verifiable)
- `python3 leads/validate_leads.py` passing before anything counts

I run this live in a harness — in the same repo you will work in.

No fallback video: this exact exercise was **tested agentically**, end to end.

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Walkthrough · Step 1 — The repo is the contract

```
marketing-research/                  github.com/tobias-weiss-ai-xr/marketing-research
├─ papers.yaml                       14,067 papers — single source of truth
├─ config/taxonomy.yaml              categories + discovery queries
├─ scripts/validate_papers.py        the contract, executable
├─ leads/leads.yaml                 your starting point: schema + rubric
├─ leads/validate_leads.py          same pattern: validate before commit
└─ README.md · statistics.json       auto-generated — never hand-edited
```

> No prose rules file the agent can ignore: **structure + validators + one source of truth.**

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Walkthrough · Step 2 — The agent at work

Watch the six-step loop run live:

- **only watch** — no magic, every tick is visible in the terminal
- agent reads the **repo** (schema + rubric), then **opens the browser** (MCP) — real company pages
- **why a browser:** LinkedIn, consent walls, logged-in portals block plain HTTP — the good context sits behind them
- writes leads into `leads/leads.yaml` → runs the **validator** → fixes what fails
- **iterates**: pause, correct, re-check — that is the loop

> Nothing happens off-screen — the loop is fully visible.

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Walkthrough · Step 3 — The three questions

1. **Checked?** Did the agent verify its output, not just generate it?
2. **Citable?** Does every claim have a URL that resolves — and actually supports it?
3. **Repeatable?** Same repo, same rubric, same validator → a result you can re-run and audit?

The 1–5 scores are a transparent heuristic — each criterion with evidence — not a measurement.

> All three guard one thing: **context** — invented context (Checked) · untraceable context (Citable) · lost context (Repeatable).

> The final slide of the walkthrough: apply them to the live result.

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Your turn: ingest + rate leads for the chair

1. **Fork & clone** — the repo is public; open `leads/`
2. **Ingest 2 leads** — real B2B companies, every claim with a URL
3. **Score & rank** — gates first, then the published formula
4. **Validate & submit** — validator passes → pull request

> Best submissions join the chair's real shortlist — "Sales Strategy and Execution", SS 2027.

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What the chair gets out of it

- A reproducible partner-sourcing method (not a one-off search)
- A ranked, sourced shortlist — fed by the course's best submissions
- Students who can run agentic workflows (a real skill for the CV)
- Evidence base for the **Bachelor course "Sales Strategy and Execution"** (SS 2027)

> The lecture is not the output. The shortlist is.

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Take-aways

1. Agentic = working environment, not a chatbot
2. Harness = model + tools + rules = accountable AI work
3. The repo is the contract: schema, rubric, validator — "done" is executable
4. Verify, cite, repeat — the three questions
5. **Context beats compute — the right context contract beats a bigger model**

> Next summer semester, the chair decides what this feeds: the Bachelor course "Sales Strategy and Execution".

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Resources

- **JLU HRZ API-Service** (free LLMs for JLU members) — uni-giessen.de → HRZ → KI → API-Service · model overview
- **HRZ access one-pager** (student quick start, 1 page) — `hrz-access-one-pager.md`
- **pi** — pi.dev · docs: pi.dev/docs · **OpenCode** — opencode.ai · **OpenRouter** (one key, many models) — openrouter.ai
- **Google AI student plan** — one.google.com/ai-student (1 year free for students)
- **marketing-research** (chair corpus, public) — github.com/tobias-weiss-ai-xr/marketing-research → start at **`leads/`** (exercise brief · rubric · validator)
- **skeleton-research** (forkable corpus skeleton) — github.com/tobias-weiss-ai-xr/skeleton-research
- **This deck** — `lecture/` in this corpus repo (English, 45 min)

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Thank you!

Questions?

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